

Evidence-Based Risk-Benefit Assessment

Purejit Shilajit Gummy Platinum (multi-ingredient dietary supplement)

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Product identified from label/photo: Purejit “Shilajit Gummy Platinum”, 60 gummies (net 240 g). Label highlights: “Purejit 3000 mg”, with KSM-66®, Ashwagandha, Black Musli, and Gokshura (Tribulus).

Note: Exact per-serving ingredient amounts and excipients (sugars, sweeteners, gelatin/pectin, etc.) were not fully available from public data at the time of writing; this limits precision in dose-related risk estimates.

1. Executive Summary (≤150 words)

Shilajit is a mineral-rich organic resin (Ayurvedic “rasayana”) commonly marketed for energy, stamina, and men’s health. Human trials of purified/processed shilajit extracts suggest potential benefits in middle-aged men (modest increases in testosterone/DHEA-S) and in exercise contexts (reduced fatigue-related strength loss), but the evidence base is small and often relies on proprietary extracts. The dominant safety issue is quality variability: crude or poorly controlled products can contain heavy metals and other contaminants, and this product also includes additional botanicals (ashwagandha and tribulus) that have published reports of liver and kidney injury in rare cases. Because this is a multi-ingredient supplement with unclear third-party verification, the overall risk level is best considered MODERATE: likely acceptable for short, cautious use in healthy adults if strong contaminant testing is verified, but not a “low-risk” everyday supplement for all users—especially older adults, pregnancy/breastfeeding, or people with liver/kidney conditions.

- Most credible benefits: modest androgen/hormone changes in a small RCT (purified shilajit), and improved post-fatigue strength retention in an 8-week trial (shilajit).
- Main concern: heavy metal contamination (lead, mercury, arsenic, and even thallium) reported in the literature for shilajit-type products; risk depends on sourcing and purification.
- Additional concern: ashwagandha and tribulus have rare but serious published liver injury cases; tribulus also has reports of kidney injury.
- Decision hinge: insist on current Certificate of Analysis (COA) from an ISO 17025 lab showing metals (Pb, As, Cd, Hg, Tl) and microbiology/mycotoxins below conservative limits.

2. Definition & Context

What it is

Shilajit (also known as mumijo/mumie) is a naturally occurring, tar-like exudate from high-altitude rocks, formed from the long-term decomposition of plant material. It is a complex mixture of fulvic/humic substances, dibenzo- α -pyrones, and minerals. It is not a single chemical and typically has no single CAS number; composition varies by region and processing.

Typical uses & where people encounter it

- Dietary supplements (resin, capsules, powders, gummies) marketed for energy, stamina, libido, “testosterone support,” and general vitality.
- Traditional use in Ayurvedic medicine as a “rasayana” (rejuvenating tonic).
- Often combined with other botanicals in “men’s vitality” formulas (e.g., ashwagandha, tribulus).

Exposure routes

- Ingestion (primary route).
- Dermal exposure is uncommon unless used in topical preparations; inhalation is not typical.

Regulatory context (important)

In most jurisdictions, shilajit products are regulated as dietary supplements or traditional products, not as approved medicines. In the U.S., dietary supplements are not FDA-approved before marketing; product quality and contaminant control are the manufacturer’s responsibility, and post-market enforcement occurs when problems are identified.

3. Benefits / Positive Effects (Table)

Practical benefits are listed below. Evidence quality varies; many studies are small and may use proprietary extracts. See Sources section for links.

Benefit	Why It Matters (real-world impact)	Evidence / Source
Possible increase in testosterone and DHEA-S in middle-aged men (purified shilajit)	May be relevant for symptomatic low-normal androgen status (energy, libido) under medical supervision.	Randomized, double-blind, placebo-controlled trial: 250 mg twice daily for 90 days increased total & free testosterone and DHEA-S vs placebo. [1]
Improved spermatogenic parameters in oligospermic men (processed shilajit)	Potential fertility relevance (should be clinician-managed; causes of infertility must be assessed).	Clinical trial in oligospermic men: 100 mg twice daily for 90 days reported improved sperm parameters and oxidative stress markers. [2]

Reduced fatigue-induced strength loss / improved recovery (shilajit supplementation)	May help retain strength after fatiguing exercise in trained/active people.	8-week trial (250–500 mg/day) showed better maximal strength retention post-fatigue vs placebo. [3]
Mechanistic neuroprotective rationale (fulvic substances; anti-aggregation effects on tau in vitro)	Theoretical relevance for cognitive aging; not the same as proven clinical benefit.	Review summarizes biochemical composition and in vitro/animal neuroprotective findings; clinical evidence for cognition remains limited. [4]
Stress/sleep support from ashwagandha component (if adequately dosed)	May reduce perceived stress and improve sleep in some adults.	NIH Office of Dietary Supplements summarizes multiple RCTs showing modest stress/anxiety improvements; also notes safety considerations. [5]

4. Risks / Negative Effects (Table)

Key risks reflect both shilajit quality concerns and added botanicals (ashwagandha, tribulus).

Acute vs chronic considerations are included where possible.

Risk / Effect	Potential Impact (mild / moderate / severe)	Type of Evidence	Key Source
Heavy metal contamination (Pb, As, Hg, Cd) and mycotoxins in crude/poorly purified shilajit	Severe (chronic accumulation can affect neurologic, renal, hematologic systems; fetal/infant risk).	Human/regulatory + review evidence	OPSS warning and review literature emphasize contaminant risk and need for purification/testing. [6][4]
Thallium (Tl) contamination reported in shilajit and shilajit-based supplements	Severe (highly toxic; accumulates in tissues; neurologic and renal injury possible).	Analytical study (supplement testing)	Study quantified Tl in crude shilajit and supplements; highlights need for standardized testing. [7]
Ashwagandha-associated liver injury (rare idiosyncratic cases; cholestatic/hepatocellular patterns reported)	Moderate to severe (jaundice, itching, hospitalization ; usually	Human case series + NIH resource	LiverTox and NIH ODS summarize published cases and caution in susceptible individuals. [8][5]

	resolves after stopping but can be serious).		
Tribulus (gokshura)– associated liver and kidney injury (rare case reports, including severe jaundice and AKI)	Severe in reported cases (bilirubin extremely high; kidney injury requiring care).	Human case reports + NIH resource	LiverTox and published case reports describe hepatotoxicity and nephrotoxicity. [9][10][11]
Product-to-product variability / mislabeling / adulteration risk (dietary supplement category)	Moderate (unexpected exposures; drug interactions; contaminant exposure).	Regulatory	FDA explains supplements are not pre-approved; post-market issues include contamination/adulteration . [12]
Glucose load/sugar alcohols (gummy format) and excipient intolerance	Mild to moderate (GI upset; glycemic impact for diabetics; dental risk).	Mechanistic/clinical common knowledge; dose-dependent	Depends on formulation; requires full Supplement Facts for assessment (data gap).
Drug interactions (possible): anticoagulants, antihypertensives, thyroid meds, immunosuppressants	Moderate to severe (bleeding risk, hypotension, thyroid changes).	Human/clinical pharmacology (variable)	Ashwagandha can affect thyroid function and sedation; tribulus may affect BP/glucose; monitor. [5][9]

5. Toxicity & Safety Profile

Overall: Shilajit is a complex natural mixture. In controlled trials using purified/processed extracts at 200–500 mg/day for 8–12 weeks (and 500 mg/day for 90 days), adverse events were generally limited, suggesting short-term tolerability in selected healthy adults. [1][3] However, this does not guarantee safety for all commercial products or for long-term use.

Short-term exposure (days to weeks)

- Expected issues (if they occur): gastrointestinal upset, headache, rash, dizziness; stop if symptoms appear.
- Rare but serious: allergic reactions have been reported; emergency care is needed for swelling, wheeze, or collapse. [6]
- For multi-ingredient formulas, attribution is uncertain; any new symptoms should be treated as product-related until proven otherwise.

Long-term / repeated exposure (months to years)

- Primary concern is cumulative exposure to toxic elements (e.g., lead, arsenic, mercury, thallium) if present at low but persistent levels. [7][6]
- Botanicals may cause idiosyncratic liver injury (ashwagandha) or rare hepato-renal injury (tribulus). [8][9][10]
- Long-term controlled human safety data for shilajit gummies specifically is not well established in public data.

Dose / exposure thresholds (what we can say with numbers)

- Shilajit trials: 250 mg twice daily (500 mg/day) for 90 days in healthy men. [1]
- Shilajit trials: 250–500 mg/day for 8 weeks in active adults. [3]
- Lead benchmarks: FDA's interim reference levels (food) are 2.2 µg/day for children and 8.8 µg/day for women of childbearing age; no safe level of lead exposure has been identified for children. [13]
- Elemental impurity limits used in pharmaceuticals (ICH Q3D) provide a conservative benchmark for oral permitted daily exposure (PDE) values (e.g., Pb 5 µg/day; inorganic As 15 µg/day; Cd 5 µg/day; Hg 30 µg/day). These are drug-quality standards, not supplement approvals, but useful for COA interpretation. [14][15]

Positions from major agencies (high-level)

- FDA: dietary supplements are not FDA-approved before marketing; quality and safety depend on manufacturing controls and post-market oversight. [12]
- FDA (2025): warns of heavy metal poisoning risk from certain unapproved Ayurvedic drug products; highlights accumulation and neurologic/renal risks. [16]
- NIH ODS: ashwagandha has evidence for stress/sleep benefits but includes safety cautions and references to liver injury reports. [5]
- CDC/ATSDR: toxicological profile documents thallium hazards and risk assessment methods (MRLs). [17]

Vulnerable groups

- Pregnant/breastfeeding: avoid unless a clinician explicitly approves and metals testing is excellent; fetal/infant vulnerability to heavy metals is a key concern. [16][13]
- Children/adolescents: avoid (no strong benefit justification; higher vulnerability to neurotoxicants). [13]
- Elderly and people with kidney disease: higher risk from contaminant accumulation and reduced renal reserve; avoid unless COA is strong and clinician monitors.
- People with liver disease or past supplement-induced liver injury: avoid due to ashwagandha/tribulus case reports. [8][9]
- People on thyroid medication or with hyperthyroidism: caution with ashwagandha (thyroid effects reported). [5]

6. Environmental Impact

Public environmental-impact data specific to shilajit harvesting and processing is limited. Key considerations are largely indirect:

- Extraction/harvesting from mountain rock faces may contribute to localized ecosystem disturbance if not managed.
- The larger environmental concern is contaminant transfer: shilajit can contain trace metals from geological sources; improper waste handling during processing could introduce metals into water/soil.

Data gap: peer-reviewed life-cycle or ecosystem studies on commercial shilajit harvesting are not well established in public data.

7. Who Should / Should Not Be Exposed (actionable guidance)

General healthy adults

- Potentially acceptable for short-term trial ONLY if the manufacturer provides a current COA (ISO 17025 lab) showing metals and microbiology/mycotoxins within conservative limits.
- Prefer cycling (e.g., 6–8 weeks) rather than indefinite use; stop immediately if jaundice, dark urine, itching, severe fatigue, abdominal pain, or confusion occurs.

Older adults (e.g., 60+)

- Risk-benefit is less favorable unless a clear, monitored goal exists (e.g., clinician-supervised low testosterone evaluation).
- Insist on COA including thallium, not just the common metals; start at the lowest dose and monitor blood pressure and GI tolerance.

Pregnancy / breastfeeding / trying to conceive

- Generally recommended to avoid due to contamination uncertainty and lack of robust safety trials; heavy metals can accumulate and may affect the fetus/infant. [16][13]

Liver/kidney disease, hypertension, diabetes, thyroid disease

- Liver disease: avoid (ashwagandha and tribulus have published liver injury cases). [8][9][10]
- Kidney disease: avoid or use only with clinician and strong COA due to metal accumulation risk. [7]
- Hypertension/diabetes: caution; potential BP/glucose effects are possible and require monitoring.

Workers with high exposure

Not typically an occupational exposure issue unless manufacturing/packaging: standard GMP controls, dust control for powders, gloves, and routine contaminant testing are appropriate.

8. Safer Alternatives or Mitigation Strategies

Mitigation if choosing to use this product

- Demand a COA for the exact batch number, dated within the last 12 months, from an ISO 17025 lab.
- COA should include: Pb, As (inorganic if possible), Cd, Hg, plus Tl; also microbial counts and mycotoxins where relevant.
- Prefer third-party certification programs when available (USP Verified, NSF, Informed Choice/Informed Sport) over generic “lab tested” claims.
- Avoid stacking with other hepatotoxic supplements (e.g., high-dose green tea extract, kava) and avoid alcohol excess during a trial period.
- Consider baseline and 4–6 week follow-up liver enzymes (ALT/AST/ALP/bilirubin) in higher-risk individuals.

Alternatives (often better-supported and lower contamination risk)

- For strength/performance: creatine monohydrate has stronger evidence and established safety for most adults (assuming medical suitability).
- For stress/sleep: ashwagandha alone from a reputable, tested brand may be easier to dose and monitor than multi-ingredient blends; still requires liver-risk awareness. [5][8]
- For low testosterone symptoms: clinician-led evaluation (sleep, weight, medications, endocrine workup) is more reliable than “testosterone booster” supplements.
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9. Comparison (Mini Table)

Option	Pros	Cons	Regulatory / Safety Notes	Best For
This multi-ingredient shilajit gummy (shilajit + ashwagandha + tribulus + “black musli”)	Convenient; targets energy/libido/stress in one product.	Harder to attribute side effects; greater interaction and liver-risk complexity; gummy excipients; needs rigorous contaminant testing.	Dietary supplement; not pre-approved; quality varies. [12]	Short, cautious trial in healthy adults only, with strong COA.
Single-ingredient purified shilajit capsule/resin from a verified third-party tested brand	Simpler exposure; easier to interpret dosing and side effects; can request COA focused on shilajit.	Still at risk of metals if poorly sourced; benefits evidence still limited.	Prefer ISO lab COA and/or recognized certification; verify metals. [6][7]	People wanting to test shilajit specifically.
Ashwagandha-only product (standardized extract) with strong QA	Better studied for stress/sleep; dosing more predictable.	Rare liver injury cases; thyroid effects possible.	NIH ODS + LiverTox summarize safety notes. [5][8]	Stress/sleep goals; avoid in liver disease.
Creatine monohydrate (performance alternative)	Strong evidence for strength/power and muscle mass; generally well studied.	May cause GI upset or water retention; caution in kidney disease; needs hydration.	Evidence base in sports nutrition literature (not covered in depth in this report).	Performance-focused use where contamination risk needs to be low.

10. Sources (official / peer-reviewed)

1. [1] Pandit S. et al. Clinical evaluation of purified Shilajit on testosterone levels in healthy volunteers. *Andrologia* (2016). PubMed: <https://pubmed.ncbi.nlm.nih.gov/26395129/>
2. [2] Biswas TK. et al. Clinical evaluation of spermatogenic activity of processed Shilajit in oligospermia. *Andrologia* (2010). PubMed: <https://pubmed.ncbi.nlm.nih.gov/20078516/>
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6. [6] Operation Supplement Safety (DoD). Shilajit as a dietary supplement ingredient (accessed Dec 2025). <https://www.opss.org/article/shilajit-dietary-supplement-ingredient>
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8. [8] NIH LiverTox. Ashwagandha (accessed Dec 2025). <https://www.ncbi.nlm.nih.gov/books/NBK548536/>
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13. [13] U.S. FDA. Guidance for Industry: Action Levels for Lead in Processed Food Intended for Babies and Young Children (2025 PDF). <https://www.fda.gov/media/164684/download>
14. [14] ICH Q3D(R2) Guideline for Elemental Impurities (2022). PDF: https://database.ich.org/sites/default/files/Q3D-R2_Guideline_Step4_2022_0308.pdf
15. [15] U.S. FDA. Q3D(R2) Elemental Impurities Guidance for Industry (2022 PDF). <https://www.fda.gov/media/148474/download>
16. [16] U.S. FDA. Heavy metal poisoning associated with certain unapproved Ayurvedic drug products (Dec 2025). <https://www.fda.gov/drugs/fraudulent-products/fda-warns-about-heavy-metal-poisoning-associated-certain-unapproved-ayurvedic-drug-products>
17. [17] ATSDR/CDC. Toxicological Profile for Thallium (Draft, 2024 PDF). <https://www.atsdr.cdc.gov/toxprofiles/tp54-a.pdf>
18. [18] Purejit product page (for context on dosing/claims; not independent evidence). <https://purejit.com/products/gummy-gold>

19. [19] Purejit “Lab testing & Certificates” page (claims of testing; COA availability unclear).
<https://purejit.com/pages/lab-testing-certificates>

Data gaps (important):

- Batch-specific COA with metals (including thallium), microbiology, and mycotoxins for the exact product batch is not publicly available in a verifiable way from the sources accessed.
- Exact per-serving dosages for ashwagandha, tribulus, and “black musli” in this gummy are not confirmed; risk depends heavily on dose and extract type.
- Long-term (multi-year) human safety data for routine shilajit use, especially in older adults, is limited.